

Beamer⁺

Interactive Presentations

a guided tour of what it can do

press → to begin

what's inside

Beamer⁺ turns a static PDF into a live, interactive lecture
every tile below has a demo later in this deck, so press → to explore



video & audio



3D models



split view



reveal steps



ink & shapes



magnifier



live widgets

```
>>>_
```

Python



live polls



scratch paper



screen recording

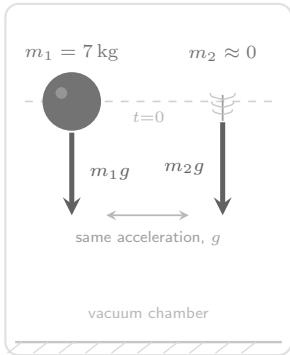


notes PDF

videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2nd law in a Physics course



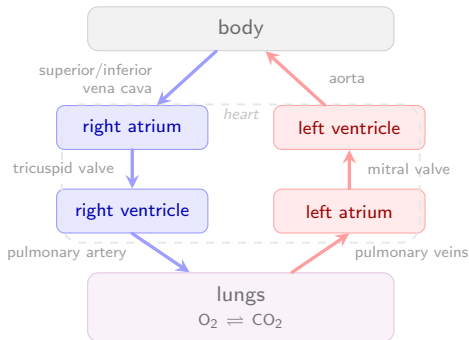
gravitational acceleration is
independent of mass



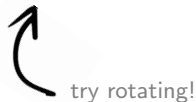
models

supported formats include .glp**b**

example usage: explaining blood flow through the heart in an Anatomy course



blood flow through the heart
(deoxygenated blood / oxygenated blood)



reveal steps

uncover a slide one piece at a time, with no \pause or recompiling
example usage: walking through a proof in a Discrete Math class

Claim. $\sqrt{2}$ is irrational.

- 1 Suppose not: $\sqrt{2} = \frac{p}{q}$ with p, q integers in *lowest terms*.
- 2 Squaring, $p^2 = 2q^2$, so p^2 is even $\Rightarrow p$ is even: $p = 2k$.
- 3 Then $4k^2 = 2q^2$, so $q^2 = 2k^2 \Rightarrow q$ is even too.
- 4 So p and q are both even, and share a factor of 2.

Contradiction! Lowest terms can't both be even. ■

 press → to reveal!

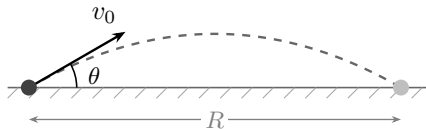
ink & annotations

pens, highlighters, shapes, $\text{\LaTeX}$ text boxes, a spotlight and a magnifier
example usage: working a projectile problem live in a Physics class

A ball is kicked at $v_0 = 20 \text{ m/s}$, at $\theta = 30^\circ$ above level ground. How far away does it land?

The range is

$$R = \frac{v_0^2 \sin 2\theta}{g} = \frac{(20)^2 \sin 60^\circ}{9.81}$$



psst, you found the fine print! Scroll while the magnifier is on to zoom anywhere from 1.5 $\times$ to 5 $\times$. It works on ink, videos and widgets too.

 try the magnifier on the fine print!

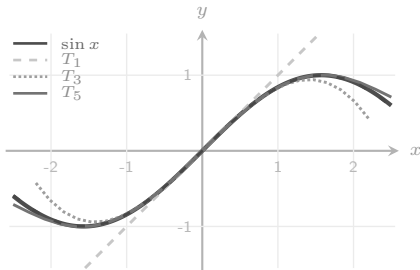
split view

allows two slides to be shown simultaneously

example usage: explaining Taylor series in a Calculus class

Any smooth function f can be approximated around $x_0 = 0$ as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



expansion of $f(x) = \sin(x)$ around $x_0 = 0$

function plotter

2D curves and 3D surfaces you can rotate, zoom and edit live
example usage: classifying critical points in a Multivariable Calculus class

At a critical point, $\nabla f = \mathbf{0}$. The
second-derivative test looks at

$$D = f_{xx}f_{yy} - f_{xy}^2.$$

For $f(x, y) = ax^2 - y^2$ at the origin:

$$D = (2a)(-2) - 0^2 = -4a$$

$a > 0$: a **saddle point**

$a < 0$: a local maximum

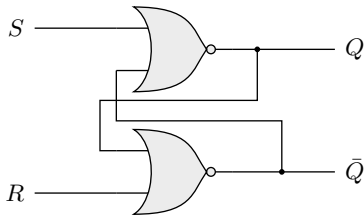
 try dragging the slider for a !

widgets

supported formats include `.html`

example usage: explaining latches in a Computer Organization class

A **latch** stores one bit of state using cross-coupled gate feedback.



cross-coupled NOR gates
inputs are *active-high*

S	R	Q	$\bar{Q}$	state
0	0	Q	$\bar{Q}$	hold
1	0	1	0	set
0	1	0	1	reset
1	1	0	0	forbidden

truth table for an SR latch
implemented with NOR gates

record & share

the lecture doesn't have to stay in the room

example usage: posting the recording and annotated notes right after class



record the tab

slides, ink, widgets
and sound, saved
every second so a
crash loses nothing



notes PDF

one handout with
your ink, every wid-
get as it stands,
and the poll results



one ZIP file

slides, media, widgets,
ink and poll setup
together, ready to
reopen next term

also: bookmarks · hide the slide with . · duplicate a slide, widgets and all

try the record button!



make your own

drop a PDF of your slides onto the window to get started

- 1 load a PDF, or a Beamer⁺ ZIP you saved earlier
- 2 open edit mode to add videos, 3D models, widgets and polls
- 3 present, draw, record, then download the ZIP and notes